Assignment No. 6

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Program

```
#include <stdio.h>
#include <ctype.h>

#define MAX 100

char stack[MAX];
int top = -1;

void push(char ch)
{
    stack[++top] = ch;
}

char pop()
{
    return stack[top--];
}

int precedence(char ch)
{
    if (ch == '^')
        return 3;
    else if (ch == '*' || ch == '/')
        return 2;
    else if (ch == '+' || ch == '-')
        return 1;
    else
        return 0;
}

void infixToPostfix(char infix[], char postfix[])
{
    int i, j = 0;
    char ch;

    top = -1;

    for (i = 0; infix[i] != '\0'; i++)
    {
        ch = infix[i];

        if (isalnum(ch))
        {
            postfix[j++] = ch;
        }

        else if (ch == '(')
        {
            push(ch);
        }

        else if (ch == ')')
        {
            while (stack[top] != '(')
            {
                postfix[j++] = pop();
            }

            pop();
        }

        else
        {
            while (top != -1 &&
                   stack[top] != '(' &&
```

```
                precedence(stack[top]) >= precedence(ch))
            {
                postfix[j++] = pop();
            }

            push(ch);
        }
    }

    while (top != -1)
    {
        postfix[j++] = pop();
    }

    postfix[j] = '\0';
}

int evaluatePostfix(char postfix[])
{
    int value[MAX];
    int vtop = -1;

    int i, a, b;
    char ch;
    int values[26] = {0};

    for (i = 0; postfix[i] != '\0'; i++)
    {
        if (isalpha(postfix[i]))
        {
            ch = postfix[i];

            if (values[ch - 'a'] == 0)
            {
                printf("Enter value of %c: ", ch);
                scanf("%d", &values[ch - 'a']);
            }
        }
    }

    for (i = 0; postfix[i] != '\0'; i++)
    {
        ch = postfix[i];

        if (isalpha(ch))
        {
            value[++vtop] = values[ch - 'a'];
        }
        else
        {
            b = value[vtop--];
            a = value[vtop--];

            switch (ch)
            {
                case '+':
                    value[++vtop] = a + b;
                    break;

                case '-':
                    value[++vtop] = a - b;
                    break;

                case '*':
                    value[++vtop] = a * b;
                    break;

                case '/':
                    value[++vtop] = a / b;
                    break;

                case '^':
                    value[++vtop] = 1;
                    break;
            }
        }
    }

    return value[vtop];
}
```

```
int main()
{
    char infix[MAX];
    char postfix[MAX];
    int result;

    printf("Enter infix expression: ");
    scanf("%s", infix);

    infixToPostfix(infix, postfix);

    printf("\nPostfix expression: %s", postfix);

    result = evaluatePostfix(postfix);

    printf("\nResult = %d\n", result);

    return 0;
}
```

Output

```
Enter infix expression: (a+b)*c

Postfix expression: ab+c*
Enter value of a: 2
Enter value of b: 3
Enter value of c: 4

Result = 20
```